

Adarsh Divase

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Professional Summary

AI Full Stack Developer with production experience designing and operating six enterprise platforms at Allwave AV Systems, covering presales, support, HR, inventory, 3D design, and AV programming. Proficient in TypeScript, Python, AWS (Amplify Gen 2, App Runner, Lambda, DynamoDB, AppSync), Gemini RAG pipelines, and LangChain. Skilled in MLOps, deep learning (CNNs, LSTMs, Transformers), and serverless architecture at scale. AWS Certified Solutions Architect with a consistent track record of shipping AI-integrated systems that solve real business problems in production.

Education

A. C. Patil College of Engineering

B.Tech in Artificial Intelligence & Data Science — GPA: 8.7 / 10.0

Navi Mumbai, India

Aug 2021 – May 2025

- Relevant Coursework: Deep Learning, Machine Learning, Data Structures & Algorithms, Distributed Database Systems, Cybersecurity

Certifications

AWS Certified Solutions Architect – Associate — Applied daily across Amplify Gen 2, Lambda, App Runner, DynamoDB, Cognito, WAF, and CDK in six production systems.

Data Engineering with Hadoop & Spark — Large-scale ETL pipeline design, distributed computing, and stream processing with Apache Kafka.

Mastering Generative AI & LLMs — Production RAG systems, async Gemini job queues, prompt engineering, and OpenAI/Gemini API integrations.

Experience

AI Full Stack Developer

Allwave AV Systems Pvt Ltd

Sep 2025 – Present

Mumbai, Maharashtra

- Architected and shipped **six production platforms** across presales, AV programming, support, HR, inventory, and 3D room design — used daily by enterprise teams across the entire organization.
- Built full-stack systems with **TypeScript/React 18** frontends and **Python/FastAPI** backends on **AWS Amplify Gen 2, App Runner, Lambda, DynamoDB, Cognito, SES, and AppSync/GraphQL**.
- Integrated **Gemini 2.5 Pro** and **LangChain RAG** pipelines grounded in curated business data; implemented async Gemini job queues via EventBridge to handle API Gateway timeout constraints.
- Owned multi-tenant **RBAC** architecture, **CDK/CI/CD** pipelines (GitHub Actions → CodeBuild → ECR → App Runner), CloudWatch/CloudTrail monitoring, and India compliance (GST, MSME, TDS, Tally XML).

Python Backend Developer Intern

NDTR Reflection Engineers Pvt Ltd

Oct 2024 – Mar 2025

Mumbai, Maharashtra

- Built scalable backend microservices with **Python, FastAPI, and Flask**, improving API response times by **25%** and maintaining **99.9% uptime** under high daily request volumes.
- Optimized **PostgreSQL** queries via **SQLAlchemy ORM** reducing retrieval latency by **15%**; secured 50+ endpoints with **JWT authentication**; integrated **SendGrid** (1,000+ emails/day) and **Razorpay**.
- Containerized all services with **Docker**, deploying to Render and Heroku and cutting deployment cycle time by **40%**.

Industry Projects

Production systems designed, built, and actively operated at **Allwave AV Systems Pvt Ltd**.

AV Inventory Ops · TypeScript 5.7 · Amplify Gen 2 · AppSync GraphQL · 36× DynamoDB · 19× Lambda · CDK · React 18 PWA · OpenSearch

- Architected a **flagship serverless back-office platform** that **replaced all spreadsheets** across inventory, procurement, billing, AMC/service contracts, and client portal — becoming the **single auditable system of record** for the entire organization and eliminating manual data reconciliation.
- Designed **36 DynamoDB tables** (8 GSIs) and **19 Lambda functions** including alert-engine, invoice-scheduler, MSME compliance checker, depreciation engine, FY rollover, TDS auto-creator, and Tally XML export generator — **automating workflows that previously required dedicated finance team effort**.
- Automated **India tax compliance** end-to-end: GST, MSME, TDS workflows, HSN validation, and 22 transactional SES HTML email templates — **reducing compliance preparation time to near-zero** per filing cycle; enforced PITR, WAF, DKIM/SPF/DMARC, and Secrets Manager across all production tables.
- Built floating **Gemini AI assistant**, Cmd+K search, real-time **OpenSearch** indexing, QR scan, and idle session monitor within a React 18 PWA — giving the operations team **instant cross-table search** over thousands of inventory and transaction records.

BINGO — Bill of Quantities Intelligent Next-Gen Optimizer v2.1 *TypeScript · Amplify Gen 2 · Gemini 2.5 Pro · LangChain RAG · DXF/CAD · ExcelJS · jsPDF · docx*

- Built a **presales AI platform** that cut **BOQ turnaround from 2–3 days to 2–3 hours** — enabling the D&E team to submit production-ready, AVIXA CTS-D compliant proposals the same day as client briefings, from a curated catalog of 2,000+ AV products.
- Engineered **Ask BINGO RAG**: vector + legacy DB search with IndexedDB caching and exact-model matching to **eliminate SKU hallucinations**; implemented AVIXA physics engines (DMD viewing distance, ACE audio coverage, VIP camera FOV) that **replaced manual engineering calculations entirely**.
- Generated full **CAD output packs** including DXF signal-flow diagrams (27+ layers), rack elevations, and cable schedules — deliverables that previously required a dedicated draftsman; deployed **async Gemini job queue** via EventBridge to bypass API Gateway timeouts.
- Enforced per-category brand rules, Cognito group RBAC, zero client-side API keys, and DynamoDB ActivityLog; **exported client-ready packs as Excel, PDF, and DOCX** eliminating all post-estimation manual formatting.

PRONTO — L1/L2 Support Copilot *Python · FastAPI · LangChain · FAISS · Gemini · Jira REST API · Lambda · Amplify*

- Developed a **production RAG support copilot** using LangChain + FAISS (S3-synced knowledge base) that **measurably reduced MTTR** on AV support tickets by surfacing cited, resolution-ready answers — cutting the time engineers spend researching known issues on repeat incidents.
- Integrated **Jira REST API** for live ticket intelligence — **AI auto-posts structured [PRONTO - AI Analysis] comments** with applied solutions directly on incidents, reducing L1 triage effort and enabling faster escalation decisions; shipped bilingual **Hindi/English voice interface** for field engineers.
- Built admin analytics dashboard and browser alerts for high-priority incidents, giving support leads **real-time visibility into SLA-critical tickets**; lazy-loaded RAGEngine, JiraService, and EscalationService to stay within Lambda cold-start INIT limits in a fully serverless deployment.

EZ Configurator — Browser 3D Room Design Studio *React 18 · Three.js · React Three Fiber · Zustand · Express · Gemini · jsPDF · Framer Motion*

- Engineered a **browser-based 3D AV design tool** with PBR rendering (6 wall absorption coefficients), SSAO/Bloom/Vignette post-processing, and snap placement across 42 SKUs from 14 brands — enabling **physics-validated room designs before purchase order**, eliminating costly on-site rework from under-specified installs.
- Implemented **AVIXA compliance engines** directly in-browser: DISCAS viewing distance, SPL acoustics ($L_p = L_w - 20\log_{10}r - 11$), Sabine RT60, PAG/NAG, and cable limits for HDMI, HDBaseT, Dante, and AVoIP — **replacing manual compliance checks** that previously required a senior AV engineer.
- Generated **complete PDF handover packs** (AVIXA compliance report + GST BOQ + IP/network scheme) via Gemini auto-design, **delivering client-ready documentation in one click**; secured all API keys server-side via Express sidecar ensuring zero credential exposure in UAT and prod.

Personal Projects

Hybrid Predictive Maintenance System — *TensorFlow, Keras, Reinforcement Learning, Streamlit, SHAP/LIME*

- Developed an **LSTM deep learning model** predicting machine health and Remaining Useful Life (RUL) from time-series sensor data, achieving a **30% improvement in prediction accuracy**.
- Combined supervised LSTM with **reinforcement learning** for optimal maintenance scheduling; SHAP/LIME dashboards reduced estimated downtime costs by **20% (\$50K/year)**.

Customer Churn Prediction API — *FastAPI, Scikit-learn, SMOTE, Uvicorn, Pandas*

- Trained a churn model achieving **F1-score 0.87**, identifying **75% of at-risk customers** within 30 days; applied SMOTE to boost minority-class recall by **25%**.
- Deployed a **RESTful inference API** via FastAPI/Uvicorn at **sub-100ms latency** sustaining **500 requests/second** throughput.

Technical Skills

Languages & Frameworks: Python, TypeScript 5.7, SQL, FastAPI, Flask, React 18, Node.js, Express, Zod, Pydantic

Cloud & DevOps (AWS): Amplify Gen 2, Lambda, DynamoDB, App Runner, Cognito, SES/SESv2, AppSync, EventBridge, Scheduler, S3, SNS, OpenSearch, Secrets Manager, WAF, ECR, CodeBuild, CloudWatch, CloudTrail, CDK, CI/CD, GitHub Actions

Generative AI & LLMs: Gemini 2.5 Pro, LangChain, RAG Pipelines, FAISS, OpenAI API, Hugging Face Transformers, Prompt Engineering, Async Job Queues, Vector Search, IndexedDB Caching

Machine Learning: Scikit-learn, XGBoost, Random Forests, SVMs, SMOTE, Feature Engineering, Model Evaluation, A/B Testing, Monte Carlo Simulation

Deep Learning: TensorFlow, PyTorch, Keras, CNNs, RNNs (LSTMs, GRUs), Transformers, GANs, Transfer Learning, Reinforcement Learning

3D Graphics & Frontend: Three.js, React Three Fiber, Drei, SSAO/Bloom post-processing, PBR rendering, Zustand, TanStack Query/Virtual, shadcn/ui, Radix UI, Tailwind CSS, Framer Motion, Recharts

Databases: DynamoDB (GSIs, PITR, streams), PostgreSQL, MongoDB, MySQL, SQLite3, SQLAlchemy ORM, OpenSearch

Data & Visualization: Plotly, Matplotlib, Seaborn, Streamlit, Dash, Prophet, SHAP, LIME, Pandas, NumPy

Specialized Domain: AVIXA CTS-D Compliance, GST/MSME/TDS Automation, Tally XML, DXF/CAD Generation (27+ layers), Jira REST API, ADF Formatting, DKIM/SPF/DMARC, Playwright E2E, Vitest